Mock S(A)MC - Created By Kevin Hu 15/03/2026

Time allowed: 90 minutes. Maximum Marks: 150. Do not worry if you cannot complete all the questions, the difficulty will increase significantly as you approach the later questions. Unattempted question = 1.5 pts, Incorrect answer = 0 pts (to discourage guessing), Correct answer = 6 pts.

1. What is the value of $\frac{2026^2-2025^2}{20.26+20.25}$?

   A 4051 B 2025.5 C 100 D 10 E 1

2. What is the smallest positive integer n such that $n!$ is divisible by 2026?

   A 503 B 1013 C 2026 D 1012 E 507

3. A sequence is defined by $a_1 = 1$ and $a_{n+1} = \frac{1}{1+a_n}$ for $n \geq 1$. What is the value of a_5?

   A $\frac{3}{5}$ B $\frac{5}{8}$ C $\frac{8}{13}$ D $\frac{2}{3}$ E $\frac{5}{7}$

4. A regular hexagon and an equilateral triangle have the same perimeter. What is the ratio of the area of the hexagon to the area of the triangle?

   A 2 : 3 B 3 : 2 C 4 : 3 D 1 : 1 E 3 : 4

5. A circle of radius 2 is tangent to both the x-axis and the y-axis. A smaller circle is tangent to the large circle and both axes. What is its radius?

   A $6-4\sqrt{2}$ B $2\sqrt{2}-2$ C $4-2\sqrt{2}$ D $1/2$ E $3-2\sqrt{2}$

6. How many integers n satisfy the inequality $n^4 - 20n^2 + 64 < 0$?

   A 2 B 4 C 6 D 8 E 10

7. The sum of the digits of a three-digit number N is 15. When reversed, the number increases by 495. How many such numbers N exist?

   A 0 B 1 C 2 D 3 E 4

8. In triangle ABC, $AB = 6, BC = 8, AC = 10$. Point P inside has distances 2 and 3 to AB and BC. What is the distance from P to AC?

   A 0.6 B 0.8 C 1 D 1.2 E 1.4

9. Let $f(x) = \frac{4^x}{4^x+2}$. Find $f(\frac{1}{2026}) + f(\frac{2}{2026}) + \cdots + f(\frac{2025}{2026})$.

   A 1012 B 1012.5 C 1013 D 2025 E 2026

10. Kevin rolls two dice. Die A has faces $\{1, 2, 2, 3, 3, 4\}$. Die B has faces $\{1, 3, 4, 5, 6, 8\}$. What is the probability that the sum of the numbers is 5?

    A 1/12 B 1/9 C 5/36 D 1/6 E 7/36

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11. For acute angles A and B, $\sin^2 A + \cos^2 B = \frac{5}{4}t$ and $\cos^2 A + \sin^2 B = \frac{3}{4}t^2$. Find the sum of all possible real t.

    A 1 B 11/3 C 5/3 D 8/3 E 4/3

12. How many positive integer pairs (x, y) satisfy $x^2 + 2xy + 3y^2 = 34$?

    A 1 B 2 C 3 D 4 E 0

13. Given real numbers a, b such that $a^2 + b^2 = 1$, what is the minimum value of $a^4 + ab + b^4$?

    A 0 B 1 C $-\frac{1}{8}$ D $\frac{9}{8}$ E $-\frac{1}{4}$

14. $ABCD$ is a unit square. E is on BC extended such that $CE = 1$. AE and DC intersect at F. BF is extended to intersect DE at G. Compute the length BG.

    A $\frac{\sqrt{6}}{3}$ B $\frac{\sqrt{5}}{3}$ C $\frac{2\sqrt{6}}{3}$ D $\frac{2\sqrt{5}}{3}$ E 2

15. Let $f\colon \mathbb{R} \to \mathbb{R}$ be the function:

$$f(x) = \sum_{k=0}^{2017} |x - k|$$

    Find the minimum value of $f(x)$ for $x \in \mathbb{R}$.

    A 1017081 B 1018081 C 2035153 D 1018080 E 1018081.5

16. Let $S = \sum_{n=1}^{2017} \frac{1}{n^3}$. Find the integer part $\lfloor 4S \rfloor$.

    A 3 B 4 C 5 D 6 E 7

17. Given the equation $4x^4 + 3x^3 - 2x^2 + 3x + 4 = 0$, find the mean of all possible values of $x + \frac{1}{x}$.

    A $-\frac{1}{4}$ B $-\frac{3}{8}$ C $\frac{5}{8}$ D 2 E $\frac{1}{8}$

18. If $x + \frac{1}{x} = \sqrt{5}$, what is the value of $x^{11} + \frac{1}{x^{11}}$?

    A $5\sqrt{5}$ B $25\sqrt{5}$ C $89\sqrt{5}$ D $123\sqrt{5}$ E $144\sqrt{5}$

19. Let $P(x)$ be a polynomial of degree n with $P(k) = \frac{k}{k+1}$ for $k = 0, \ldots, n$. Find $P(n+1)$.

    A $\frac{n+1}{n+2}$ B 1 C $\frac{n+1+(-1)^{n+1}}{n+2}$ D $\frac{n}{n+2}$ E $\frac{n+1+(-1)^n}{n+2}$

20. Isosceles triangle ABC has $\angle A = 20°, \angle B = \angle C = 80°$. D is on AB such that $AD = BC$. Compute $\angle BDC$.

    A 20° B 25° C 30° D 35° E 40°

21. If a, b, c are roots of $x^3 - 5x^2 + 7x - 9 = 0$, find $\frac{a^2}{bc} + \frac{b^2}{ac} + \frac{c^2}{ab}$.

    A 46/9 B 47/9 C 16/3 D 49/9 E 5

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22. The number $N = 11\ldots1$ (81 ones) is divisible by which of the following?

A 81 B 243 C 6561 D 3^9 E 9^9

23. If the equation $x^2 + 2px - 3p - 2 = 0$ has two distinct real roots x_1, x_2 such that

$$x_1^2 + x_1^3 = 4 - (x_2^2 + x_2^3)$$

Find the sum of all possible real values of p.

A 0 B $-\frac{3}{4}$ C -1 D $-\frac{5}{4}$ E $-\frac{3}{2}$

24. Let a, b, c be real numbers such that $abc = -1$ and $a + b + c = 4$. Given that

$$\sum_{cyc} \frac{a}{a^2 - 3a - 1} = \frac{4}{9}$$

where the sum is taken over the cyclic permutations of (a, b, c), find the value of

$$a^2 + b^2 + c^2$$

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A 15 B 9 C 33/4 D 8 E 33/2

25. In $\triangle ABC$, the $\angle BAC = 60^\circ$ and the side lengths satisfy $AB = 2AC$. Let P be a point inside $\triangle ABC$ such that:

$$PA = \sqrt{3}, \quad PB = 5, \quad PC = 2$$

Compute the area of $\triangle ABC$.

A $\frac{7\sqrt{3}+6}{2}$ B $\frac{10\sqrt{3}+8}{3}$ C $5 + 2\sqrt{3}$ D $7\sqrt{3} - 2$ E $\frac{13\sqrt{3}}{4}$

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